at any time t, has the partial information $\mathbf{x}_t$ fully available and can progressively estimate:

$$\mathbf{x}_0 \approx \hat{\mathbf{x}}_0 = \left(\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t)\right)/\sqrt{\bar{\alpha}_t} \tag{15}$$

due to Eq. (4). (A stochastic reconstruction $\mathbf{x}_0 \sim p_\theta(\mathbf{x}_0|\mathbf{x}_t)$ is also valid, but we do not consider it here because it makes distortion more difficult to evaluate.) Figure 5 shows the resulting rate-distortion plot on the CIFAR10 test set. At each time t, the distortion is calculated as the root mean squared error $\sqrt{\|\mathbf{x}_0 - \hat{\mathbf{x}}_0\|^2/D}$, and the rate is calculated as the cumulative number of bits received so far at time t. The distortion decreases steeply in the low-rate region of the rate-distortion plot, indicating that the majority of the bits are indeed allocated to imperceptible distortions.

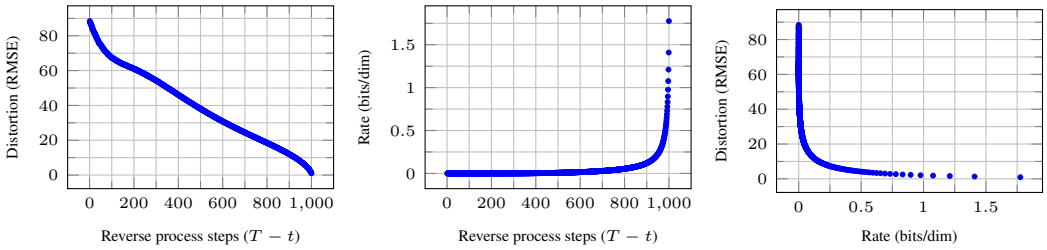

Figure 5: Unconditional CIFAR10 test set rate-distortion vs. time. Distortion is measured in root mean squared error on a $[0, 255]$ scale. See Table 4 for details.

Progressive generation We also run a progressive unconditional generation process given by progressive decompression from random bits. In other words, we predict the result of the reverse process, $\hat{\mathbf{x}}_0$, while sampling from the reverse process using Algorithm 2. Figures 6 and 10 show the resulting sample quality of $\hat{\mathbf{x}}_0$ over the course of the reverse process. Large scale image features appear first and details appear last. Figure 7 shows stochastic predictions $\mathbf{x}_0 \sim p_\theta(\mathbf{x}_0|\mathbf{x}_t)$ with $\mathbf{x}_t$ frozen for various t. When t is small, all but fine details are preserved, and when t is large, only large scale features are preserved. Perhaps these are hints of conceptual compression [18].

Figure 6: Unconditional CIFAR10 progressive generation ($\hat{\mathbf{x}}_0$ over time, from left to right). Extended samples and sample quality metrics over time in the appendix (Figs. 10 and 14).

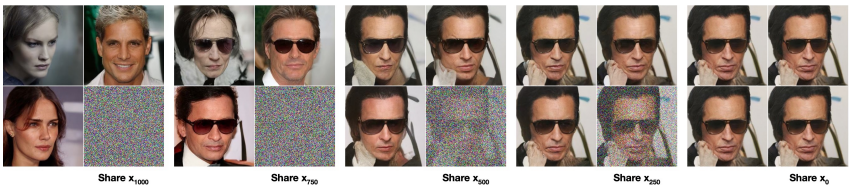

Figure 7: When conditioned on the same latent, CelebA-HQ 256×256 samples share high-level attributes. Bottom-right quadrants are $\mathbf{x}_t$, and other quadrants are samples from $p_\theta(\mathbf{x}_0|\mathbf{x}_t)$.

Connection to autoregressive decoding Note that the variational bound (5) can be rewritten as:

$$L = D_{\mathrm{KL}}(q(\mathbf{x}_T) \parallel p(\mathbf{x}_T)) + \mathbb{E}_q\left[\sum_{t\geq 1} D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))\right] + H(\mathbf{x}_0) \tag{16}$$

(See Appendix A for a derivation.) Now consider setting the diffusion process length T to the dimensionality of the data, defining the forward process so that $q(\mathbf{x}_t|\mathbf{x}_0)$ places all probability mass on $\mathbf{x}_0$ with the first t coordinates masked out (i.e. $q(\mathbf{x}_t|\mathbf{x}_{t-1})$ masks out the tth coordinate), setting $p(\mathbf{x}_T)$ to place all mass on a blank image, and, for the sake of argument, taking $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ to

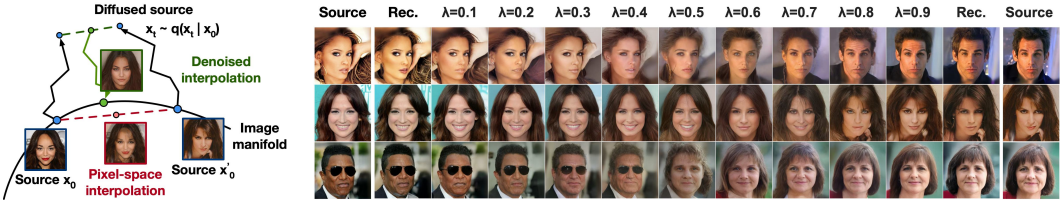

Figure 8: Interpolations of CelebA-HQ 256x256 images with 500 timesteps of diffusion.

be a fully expressive conditional distribution. With these choices, $D_{\mathrm{KL}}(q(\mathbf{x}_T) \parallel p(\mathbf{x}_T)) = 0$, and minimizing $D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))$ trains p_θ to copy coordinates $t+1, \ldots, T$ unchanged and to predict the tth coordinate given $t+1, \ldots, T$. Thus, training p_θ with this particular diffusion is training an autoregressive model.

We can therefore interpret the Gaussian diffusion model (2) as a kind of autoregressive model with a generalized bit ordering that cannot be expressed by reordering data coordinates. Prior work has shown that such reorderings introduce inductive biases that have an impact on sample quality [38], so we speculate that the Gaussian diffusion serves a similar purpose, perhaps to greater effect since Gaussian noise might be more natural to add to images compared to masking noise. Moreover, the Gaussian diffusion length is not restricted to equal the data dimension; for instance, we use $T = 1000$, which is less than the dimension of the $32 \times 32 \times 3$ or $256 \times 256 \times 3$ images in our experiments. Gaussian diffusions can be made shorter for fast sampling or longer for model expressiveness.

4.4 Interpolation

We can interpolate source images $\mathbf{x}_0, \mathbf{x}'_0 \sim q(\mathbf{x}_0)$ in latent space using q as a stochastic encoder, $\mathbf{x}_t, \mathbf{x}'_t \sim q(\mathbf{x}_t|\mathbf{x}_0)$, then decoding the linearly interpolated latent $\bar{\mathbf{x}}_t = (1-\lambda)\mathbf{x}_0 + \lambda\mathbf{x}'_0$ into image space by the reverse process, $\bar{\mathbf{x}}_0 \sim p(\mathbf{x}_0|\bar{\mathbf{x}}_t)$. In effect, we use the reverse process to remove artifacts from linearly interpolating corrupted versions of the source images, as depicted in Fig. 8 (left). We fixed the noise for different values of λ so $\mathbf{x}_t$ and $\mathbf{x}'_t$ remain the same. Fig. 8 (right) shows interpolations and reconstructions of original CelebA-HQ 256×256 images ($t = 500$). The reverse process produces high-quality reconstructions, and plausible interpolations that smoothly vary attributes such as pose, skin tone, hairstyle, expression and background, but not eyewear. Larger t results in coarser and more varied interpolations, with novel samples at $t = 1000$ (Appendix Fig. 9).

5 Related Work

While diffusion models might resemble flows [9, 46, 10, 32, 5, 16, 23] and VAEs [33, 47, 37], diffusion models are designed so that q has no parameters and the top-level latent $\mathbf{x}_T$ has nearly zero mutual information with the data $\mathbf{x}_0$. Our ϵ-prediction reverse process parameterization establishes a connection between diffusion models and denoising score matching over multiple noise levels with annealed Langevin dynamics for sampling [55, 56]. Diffusion models, however, admit straightforward log likelihood evaluation, and the training procedure explicitly trains the Langevin dynamics sampler using variational inference (see Appendix C for details). The connection also has the reverse implication that a certain weighted form of denoising score matching is the same as variational inference to train a Langevin-like sampler. Other methods for learning transition operators of Markov chains include infusion training [2], variational walkback [15], generative stochastic networks [1], and others [50, 54, 36, 42, 35, 65].

By the known connection between score matching and energy-based modeling, our work could have implications for other recent work on energy-based models [67–69, 12, 70, 13, 11, 41, 17, 8]. Our rate-distortion curves are computed over time in one evaluation of the variational bound, reminiscent of how rate-distortion curves can be computed over distortion penalties in one run of annealed importance sampling [24]. Our progressive decoding argument can be seen in convolutional DRAW and related models [18, 40] and may also lead to more general designs for subscale orderings or sampling strategies for autoregressive models [38, 64].

6 Conclusion

We have presented high quality image samples using diffusion models, and we have found connections among diffusion models and variational inference for training Markov chains, denoising score matching and annealed Langevin dynamics (and energy-based models by extension), autoregressive models, and progressive lossy compression. Since diffusion models seem to have excellent inductive biases for image data, we look forward to investigating their utility in other data modalities and as components in other types of generative models and machine learning systems.

Broader Impact

Our work on diffusion models takes on a similar scope as existing work on other types of deep generative models, such as efforts to improve the sample quality of GANs, flows, autoregressive models, and so forth. Our paper represents progress in making diffusion models a generally useful tool in this family of techniques, so it may serve to amplify any impacts that generative models have had (and will have) on the broader world.

Unfortunately, there are numerous well-known malicious uses of generative models. Sample generation techniques can be employed to produce fake images and videos of high profile figures for political purposes. While fake images were manually created long before software tools were available, generative models such as ours make the process easier. Fortunately, CNN-generated images currently have subtle flaws that allow detection [62], but improvements in generative models may make this more difficult. Generative models also reflect the biases in the datasets on which they are trained. As many large datasets are collected from the internet by automated systems, it can be difficult to remove these biases, especially when the images are unlabeled. If samples from generative models trained on these datasets proliferate throughout the internet, then these biases will only be reinforced further.

On the other hand, diffusion models may be useful for data compression, which, as data becomes higher resolution and as global internet traffic increases, might be crucial to ensure accessibility of the internet to wide audiences. Our work might contribute to representation learning on unlabeled raw data for a large range of downstream tasks, from image classification to reinforcement learning, and diffusion models might also become viable for creative uses in art, photography, and music.

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References

[1] Guillaume Alain, Yoshua Bengio, Li Yao, Jason Yosinski, Eric Thibodeau-Laufer, Saizheng Zhang, and Pascal Vincent. GSNs: generative stochastic networks. *Information and Inference: A Journal of the IMA*, 5(2):210–249, 2016.

[2] Florian Bordes, Sina Honari, and Pascal Vincent. Learning to generate samples from noise through infusion training. In *International Conference on Learning Representations*, 2017.

[3] Andrew Brock, Jeff Donahue, and Karen Simonyan. Large scale GAN training for high fidelity natural image synthesis. In *International Conference on Learning Representations*, 2019.

[4] Tong Che, Ruixiang Zhang, Jascha Sohl-Dickstein, Hugo Larochelle, Liam Paull, Yuan Cao, and Yoshua Bengio. Your GAN is secretly an energy-based model and you should use discriminator driven latent sampling. *arXiv preprint arXiv:2003.06060*, 2020.

[5] Tian Qi Chen, Yulia Rubanova, Jesse Bettencourt, and David K Duvenaud. Neural ordinary differential equations. In *Advances in Neural Information Processing Systems*, pages 6571–6583, 2018.

[6] Xi Chen, Nikhil Mishra, Mostafa Rohaninejad, and Pieter Abbeel. PixelSNAIL: An improved autoregressive generative model. In *International Conference on Machine Learning*, pages 863–871, 2018.

[7] Rewon Child, Scott Gray, Alec Radford, and Ilya Sutskever. Generating long sequences with sparse transformers. *arXiv preprint arXiv:1904.10509*, 2019.